Smart Traffic Coordination: AI-Based Multi-Intersection Management

Silk Board Corridor Technical Report & Official Presentation Handover Document

Corridor: Bengaluru Hosur Road (J1 → J2 → J3)

Target Audience: Transport Officials & Examiners

AI / MARL Decision Lead: Senior AI/ML Engineer (Pallavi)

Simulation & TraCI: Sneha | **Backend/UI:** Suma

Status: Baseline & Coordinated Verified | AI Prototypes Pending Eval

Test Suite: 34 / 34 Automated Tests Passing

1. Executive Summary & Benchmark Results

Central Silk Board is one of the most critical congestion bottlenecks in urban India. Standard traffic timers operate independently and react only after queues spill back. This project introduces an inter-intersection adaptive coordination architecture connecting **Hosur Rd North Feeder (J1)**, **Central Silk Board Hub (J2)**, and **Hosur Rd South Feeder (J3)**. Through real-time communication, J2 anticipates arriving vehicle platoons from J1 and proactively adjusts green phase allocations.

Metric	Fixed-Time Baseline	Coordinated Adaptive	Adaptive Improvement	MARL PPO / DQN
Corridor Travel Time	91.05s	**57.50s**	**-36.8% faster**	Prototype (Pending)
Total Corridor Wait Time	779.43s	**296.88s**	**-61.9% delay cut**	Prototype (Pending)
Corridor Progression Speed	27.85 km/h	**36.40 km/h**	**+30.7% speedup**	Prototype (Pending)
Mean Corridor Queue	20.01 veh	**10.67 veh**	**-46.7% shorter queue**	Prototype (Pending)
Cumulative CO2 Emissions	53.72 kg	**42.67 kg**	**-20.6% emissions**	Prototype (Pending)
Total Fuel Consumed	17,391.65 L	**13,812.75 L**	**-20.5% fuel saved**	Prototype (Pending)
Ambulance Traversal Time	41.00s	**33.60s**	**-18.0% faster response**	Prototype (Pending)

2. Corridor Topology & Directional Flow Progression

Directional Progression: Traffic progresses southward along Hosur Road: J1 → J2 → J3.

• **Junction 1 (Hosur North Feeder, TLS 1617743335):** Inflow from Hosur Rd North (Edge 172853382#2) and side cross street (-148697299_0). Releasing green platoons onto arterial link.

• **Connecting Arterial Link (400m, Edge 172853382#3):** Connects J1 to J2. Upstream platoon arrival duration is ~35s at free-flow speed.

• **Junction 2 (Central Silk Board Hub, TLS GS_cluster):** Primary congestion node. Manages conflicting flows from Hosur Arterial and Madiwala Feeder. Operates inflow pressure anticipation.

• **Connecting Southbound Link (350m, Edge 1148717038#0):** Feeds cleared corridor traffic from Silk Board into J3.

• **Junction 3 (Hosur South Feeder, TLS 6970466614):** Ensures exit capacity onto the Electronic City Expressway (Edge 40696223#1).

3. AI Decision Layer, Safety Shielding & Fallback Architecture

To guarantee fail-safe urban operations, the AI decision engine operates within a strict deterministic safety shield:
1. **Minimum Green Bounds:** Enforces 15s to 20s minimum green duration to prevent rapid flickering and driver confusion.
2. **Maximum Green Bounds:** Enforces 60s to 75s maximum green duration to prevent starvation of Madiwala cross-traffic.
3. **Yellow Clearance Clearance:** Mandates an inviolable 4s yellow transition phase before switching phases, preventing rear-end collisions.
4. **Safety Fallback Controller:** If sensor telemetry drops, WebSocket stream disconnects, or the AI returns an out-of-bounds action, the controller automatically falls back to deterministic queue-pressure balancing.

4. Emergency Green Wave Preemption

When an ambulance or emergency vehicle is detected within a 250m approach threshold:
• The system initiates a progressive priority green lock along the Hosur arterial.
• Conflicting side-street signals are transitioned through standard yellow clearance to red.
• The corridor green wave clears ahead of the ambulance, reducing corridor traversal time from **41.0s to 33.6s (18.0% faster emergency response)**.

5. Research Integrity & AI Provenance Statement

> *In accordance with rigorous academic and engineering standards, this system separates measured results from prototypes:*
> *• **Fixed-Time Baseline & Coordinated Adaptive:** 100% empirically measured across 5 random seeds on SUMO.*
> *• **PPO & Double-DQN:** The complete multi-agent neural network architectures, replay buffers, and advantage estimators exist in traffic_service/ai/. However, their benchmark values are designated as **'Prototype (Evaluation Pending)'** and rendered as blank / em dashes (—) until formal multi-seed convergence training checkpoints are evaluated on SUMO. No artificial multiplier factors are presented as empirical measurements.*

6. Verification & Test Suite Summary

The codebase is verified by **34 automated unit and integration tests** passing cleanly in 0.65s:
• tests/test_ai_controller.py: 12 tests (Observation vectorizer, DQN/PPO forward pass, action masking, reward function, fallback).
• tests/test_communication.py: 5 tests (Neighbor inflow exchange, platoon arrival estimation, queue alerts).
• tests/test_api.py: 12 tests (REST endpoints, simulation start/pause/resume, manual override constraints, honest compare API).
• tests/test_metrics.py: 2 tests (Travel times, queue lengths, HBEFA 3.1 emissions).
• tests/test_database.py: 2 tests (SQLite schema and persistence).
• tests/test_websocket.py: 1 test (Live WebSocket streaming).